

Dynamics of Modified Square Maps over p -adic Rings for Fermat Primes

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Abstract

After introducing the fundamentals of the p -adic setting and the basic language of discrete dynamical systems on $\mathbb{Z}_p$, we develop the reduction-and-lifting toolkit used to describe decompositions of $\mathbb{Z}_p$. We then study linearly perturbed square maps $f(x) = x^2 - p^r x$ over $\mathbb{Z}_p$ for Fermat primes p . Using finite reductions modulo powers of p , cycle lifting, and minimal invariant unions of clopen balls, we derive a closed-form minimal decomposition of $\mathbb{Z}_p$ for this family in the Fermat-prime case, specifying exactly which p -adic balls make up each minimal component and how these balls are indexed.

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1 Introduction

The behavior of polynomial maps on p -adic integer rings has been studied from several viewpoints [5, 1, 6, 8].

In particular, in the study of minimality, finite reductions modulo p^k are often used as an effective method for detecting minimal components. Writing the resulting decomposition with minimal components isolated, often called the *minimal decomposition* in the literature [7, 6], can be difficult. This paper focuses on the family $f(x) = x^2 - p^r x$ for Fermat primes $p = 2^{2^\ell} + 1$, where the arithmetic of multiplication by 2 modulo p allows the decomposition to be written in closed form, down to the explicit indexing of the individual p -adic balls appearing at each level.

The resulting formula applies uniformly from small examples to large systems, including, for instance,

$$(\mathbb{Z}_{257}, x^2 - 16974593x),$$

which is simply the case where $p = 257$ and $r = 3$.

2 Preliminaries

2.1 The p -adic Setting

The p -adic numbers arise from expansions in powers of a fixed prime p . In this and the following sections, we pay most of our attention to the ring of p -adic integers $\mathbb{Z}_p$ and only briefly discuss the field $\mathbb{Q}_p$ for context.

Definition 2.1. Given a prime p , a p -adic integer is a formal infinite series

$$a_0 + a_1 p + a_2 p^2 + \cdots, \quad a_i \in \{0, \dots, p-1\} \ \forall i \geq 0.$$

We denote the set of all p -adic integers by $\mathbb{Z}_p$, and the set of all p -adic numbers by $\mathbb{Q}_p = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}_p, b \neq 0 \right\}$.

Definition 2.2. The p -adic valuation is the function

$$v_p : \mathbb{Q} \rightarrow \mathbb{Z} \cup \{\infty\}$$

defined by

$$v_p(x) = \begin{cases} k & \text{if } x = p^k \frac{a}{b}, \ a, b, k \in \mathbb{Z}, \ p \nmid a, \ p \nmid b, \\ \infty & \text{if } x = 0. \end{cases}$$

Immediately from the definition, we also see that

$$\begin{aligned} v_p(xy) &= v_p(x) + v_p(y), & x, y \in \mathbb{Q}, \\ v_p\left(\frac{x}{y}\right) &= v_p(x) - v_p(y), & x \in \mathbb{Q}, \ y \in \mathbb{Q}^\times. \end{aligned}$$

Definition 2.3. The p -adic absolute value $|\cdot|_p$ of $x \in \mathbb{Q}$ is defined by

$$|x|_p = \begin{cases} p^{-v_p(x)} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

This induces the p -adic metric or, equivalently, the p -adic distance

$$d_p(x, y) = |x - y|_p.$$

Proposition 2.4. For all $x, y \in \mathbb{Q}$, $|x + y|_p \leq \max\{|x|_p, |y|_p\}$. Therefore, $|\cdot|_p$ is non-Archimedean, and the metric $d_p(x, y) = |x - y|_p$ is ultrametric.

Proof. It suffices to show that $v_p(x + y) \geq \min\{v_p(x), v_p(y)\}$. Write $x = p^k \frac{a}{b}$ and $y = p^\ell \frac{c}{d}$, where $p \nmid a, b, c, d$. Then $v_p(x) = k$ and $v_p(y) = \ell$. WLOG, assume $k \leq \ell$. Then $x + y = p^k \frac{a}{b} + p^\ell \frac{c}{d} = p^k \left(\frac{a}{b} + p^{\ell-k} \frac{c}{d} \right) = p^k \left(\frac{ad + p^{\ell-k}bc}{bd} \right)$. Since $k \leq \ell$, we have $\ell - k \geq 0$, so $p^{\ell-k} \in \mathbb{Z}$, hence $ad + p^{\ell-k}bc \in \mathbb{Z}$. Also, since $p \nmid b$ and $p \nmid d$, we have $p \nmid bd$, so $v_p(bd) = 0$. Therefore, $v_p\left(\frac{ad + p^{\ell-k}bc}{bd}\right) = v_p(ad + p^{\ell-k}bc) - v_p(bd) = v_p(ad + p^{\ell-k}bc) \geq 0$. It follows that $v_p(x + y) = k + v_p\left(\frac{ad + p^{\ell-k}bc}{bd}\right) \geq k = \min\{k, \ell\} = \min\{v_p(x), v_p(y)\}$. Thus, $|x + y|_p \leq \max\{|x|_p, |y|_p\}$. $\square$

Definition 2.5. [11] Let $U \subseteq \mathbb{Q}_p$ be open and let $f : U \rightarrow \mathbb{Q}_p$. We say that f is p -adically differentiable at $x \in U$ if the limit

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

exists in $\mathbb{Q}_p$.

Note that the limit is understood in the p -adic sense; i.e., $h \rightarrow 0$ means $|h|_p \rightarrow 0$.

Proposition 2.6. In any ultrametric space, including $\mathbb{Z}_p$ and $\mathbb{Q}_p$, every ball is clopen.

Proof. First, consider the open ball $B_r(x)$. Since open balls are open by definition, it remains to show that $B_r(x)$ is closed. Let $y \notin B_r(x)$, so $d(x, y) \geq r$. If $z \in B_r(y)$, then $d(y, z) < r$. If we had $z \in B_r(x)$, then $d(x, z) < r$, and the ultrametric inequality would give

$$d(x, y) \leq \max\{d(x, z), d(z, y)\} < r,$$

contradicting $d(x, y) \geq r$. Hence $B_r(y) \subseteq X \setminus B_r(x)$, so the complement of $B_r(x)$ is open. Therefore, $B_r(x)$ is closed.

Now consider the closed ball $\overline{B}_r(x)$. Since closed balls are closed in any metric space, it remains to show that $\overline{B}_r(x)$ is open. Let $y \in \overline{B}_r(x)$, so $d(x, y) \leq r$. If $z \in B_r(y)$, then $d(y, z) < r$, and thus

$$d(x, z) \leq \max\{d(x, y), d(y, z)\} \leq r.$$

Thus, $B_r(y) \subseteq \overline{B}_r(x)$, so every point of $\overline{B}_r(x)$ has an open neighborhood contained in $\overline{B}_r(x)$. Therefore, $\overline{B}_r(x)$ is open. Thus, we have shown that every open or closed ball in an ultrametric space is clopen. $\square$

In the following diagram, and throughout the paper, we use the coset notation

$$a + p^k \mathbb{Z}_p = \{a + p^k u : u \in \mathbb{Z}_p\}, \quad k \geq 1,$$

for the clopen p -adic ball of radius p^{-k} , equivalently the residue class of a modulo p^k inside $\mathbb{Z}_p$. In particular, for $p = 3$,

$$a + 3^k \mathbb{Z}_3 = \{a + 3^k u : u \in \mathbb{Z}_3\}.$$

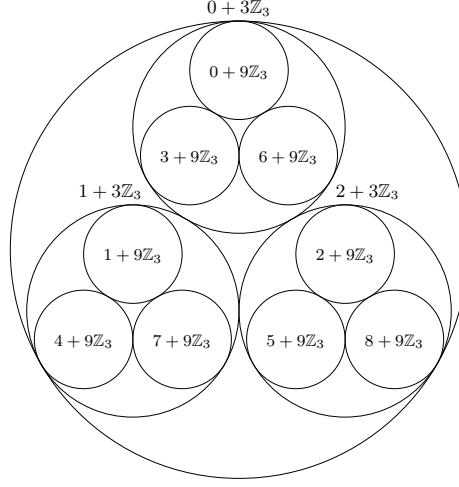


Figure 1. Nested 3-adic balls in $\mathbb{Z}_3$.

For a better intuitive understanding of the topology and other properties of $\mathbb{Q}_p$, the reader is encouraged to consult sources beyond the scope of this paper [9, 11, 10].

2.2 Discrete Dynamical Systems on $\mathbb{Z}_p$

Definition 2.7. A *discrete dynamical system* is a pair (X, f) , where X is a set and $f : X \rightarrow X$ is a self-map. For each $x \in X$, the dynamics are given by the iterates of f , namely

$$x, f(x), f^2(x), f^3(x), \dots,$$

where

$$f^n = \underbrace{f \circ f \circ \dots \circ f}_{n \text{ times}} \quad (n \geq 1), \quad \text{and} \quad f^0 = \text{id}_X.$$

Definition 2.8. Let (X, f) be a discrete dynamical system. For $x \in X$, the *orbit* of x under f is

$$\mathcal{O}_f(x) = \{f^n(x) : n \geq 0\}.$$

A point $x \in X$ is a *fixed point* of f if $f(x) = x$. More generally, x is a *periodic point* of f if there exists $n \geq 1$ such that $f^n(x) = x$.

Definition 2.9. [12] Let $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ be a p -adically differentiable map, and let $x \in \mathbb{Z}_p$ be a periodic point of least period n . Then:

1. If $|(f^n)'(x)|_p < 1$, then $\mathcal{O}_f(x)$ is said to be *attracting*.
2. If $|(f^n)'(x)|_p > 1$, then $\mathcal{O}_f(x)$ is said to be *repelling*.
3. If $|(f^n)'(x)|_p = 1$, then $\mathcal{O}_f(x)$ is said to be *indifferent*.

Definition 2.10. Let (X, f) be a discrete dynamical system. A subset $A \subseteq X$ is called *invariant* if, for every $x \in A$, $f^n(x) \in A$ for all $n \geq 0$.

Definition 2.11. Let $(\mathbb{Z}_p, f)$ be a discrete dynamical system, where $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$, and let $\mathcal{O}_f(x)$ be an attracting periodic orbit. The *basin of attraction* of $\mathcal{O}_f(x)$ is the set

$$\mathcal{B}(\mathcal{O}_f(x)) = \{y \in \mathbb{Z}_p : d_p(f^m(y), \mathcal{O}_f(x)) \rightarrow 0 \text{ as } m \rightarrow \infty\},$$

where

$$d_p(\xi, \mathcal{O}_f(x)) = \min_{a \in \mathcal{O}_f(x)} d_p(\xi, a).$$

Definition 2.12. Let (X, f) be a discrete dynamical system on a topological space X . The orbit $\mathcal{O}_f(x)$ is called *dense* in X if, for every nonempty open set $U \subseteq X$,

$$\mathcal{O}_f(x) \cap U \neq \emptyset.$$

Definition 2.13. [3] A discrete dynamical system (X, f) is called *minimal* if the orbit of every point in X is dense in X .

Proposition 2.14. Let $f \in \mathbb{Z}_p[x]$. Then f is 1-Lipschitz on $\mathbb{Z}_p$.

Proof. Write $f(x) = \sum_{j=0}^n a_j x^j$ with $a_j \in \mathbb{Z}_p$. For $x, y \in \mathbb{Z}_p$,

$$x^j - y^j = (x - y)(x^{j-1} + x^{j-2}y + \cdots + y^{j-1}),$$

so $|x^j - y^j|_p \leq |x - y|_p$. Since $|a_j|_p \leq 1$, the ultrametric inequality gives

$$|f(x) - f(y)|_p \leq \max_{0 \leq j \leq n} |a_j(x^j - y^j)|_p \leq |x - y|_p.$$

Thus, f is 1-Lipschitz. □

2.3 Decomposition, Minimality, and Cycles on $\mathbb{Z}_p$

A pair $(\mathbb{Z}_p, f)$ with $f \in \mathbb{Z}_p[x]$ defines a 1-Lipschitz polynomial dynamical system on the ring of p -adic integers. Minimality plays a central role in the study of such systems. When global minimality fails, one is naturally led to study the decomposition of $(\mathbb{Z}_p, f)$, and in particular its minimal subsystems.

Theorem 2.15. [6] Let $f \in \mathbb{Z}_p[x]$ be a polynomial of degree at least 2. We have the following decomposition

$$\mathbb{Z}_p = \mathcal{P} \sqcup \mathcal{M} \sqcup \mathcal{B},$$

where

1. $\mathcal{P}$ is the finite set consisting of all periodic points of f ;
2. $\mathcal{M} = \bigsqcup_i \mathcal{M}_i$ is the union of all at most countably many clopen invariant sets such that each $\mathcal{M}_i$ is a finite union of balls and each subsystem $f : \mathcal{M}_i \rightarrow \mathcal{M}_i$ is minimal;
3. each point in $\mathcal{B}$ lies in the basin of attraction of a periodic orbit or of a minimal subsystem.

This decomposition will be referred to as a *minimal decomposition*, and the invariant subsets $\mathcal{M}_i$ will be called *minimal components*.

Definition 2.16. Let $f \in \mathbb{Z}_p[x]$. Given $k \geq 1$, the *reduction of f modulo p^k* is the map

$$f_k : \mathbb{Z}_p/p^k\mathbb{Z}_p \rightarrow \mathbb{Z}_p/p^k\mathbb{Z}_p, \quad f_k(x + p^k\mathbb{Z}_p) = f(x) + p^k\mathbb{Z}_p.$$

One way to understand the function of reductions is to imagine them being low resolution shadows of the dynamics on $\mathbb{Z}_p$. As we increase k in p^k , the resolution of the shadow becomes more and more detailed, and so provides a progressively finer approximation of the true dynamical behavior.

Theorem 2.17. [8] Let $f \in \mathbb{Z}_p[x]$, and let $E \subseteq \mathbb{Z}_p$ be a compact f -invariant set. For each $k \geq 1$, let $f_k : E/p^k\mathbb{Z}_p \rightarrow E/p^k\mathbb{Z}_p$ be the map induced by reduction modulo p^k . Then $f : E \rightarrow E$ is minimal if and only if $f_k : E/p^k\mathbb{Z}_p \rightarrow E/p^k\mathbb{Z}_p$ is minimal for every $k \geq 1$.

Theorem 2.18. [6] Let $f \in \mathbb{Z}_p[x]$.

1. If $p \geq 5$, the periods of periodic orbits are of the form ab , where $a \mid (p-1)$ and $1 \leq b \leq p$.
2. If $p = 3$, the periods of periodic orbits must be 1, 2, 3, 4, 6, or 9.
3. If $p = 2$, the periods of periodic orbits must be 1, 2, or 4.
4. Let $p = 2$. If there is a 4-periodic orbit, then $f(z \bmod 2) \bmod 2$ should be a permutation on $\mathbb{Z}/2\mathbb{Z}$. There is no 4-periodic orbit for quadratic polynomials.

Definition 2.19. Let $f \in \mathbb{Z}_p[x]$ and $k \geq 1$. A cycle of length ℓ (also called an ℓ -cycle) modulo p^k is a tuple

$$\sigma = (x_1, \dots, x_\ell)$$

of distinct elements of $\mathbb{Z}/p^k\mathbb{Z}$ such that

$$f_k(x_i) = x_{i+1} \ (1 \leq i < \ell), \quad f_k(x_\ell) = x_1.$$

We use cycles as one of the primary tools to investigate the behavior of a given p -adic dynamical system. In particular, for odd primes p , cycles can *grow*, *split*, or *grow tails*: this classification allows us to predict the behavior of the system after the so-called *cycle lifting*. We now proceed to derive a function that will describe precisely that.

Let $f \in \mathbb{Z}_p[x]$, and let $\sigma = (x_1, \dots, x_\ell)$ be an ℓ -cycle of the induced map $f_k : \mathbb{Z}_p/p^k\mathbb{Z}_p \rightarrow \mathbb{Z}_p/p^k\mathbb{Z}_p$. Set $g = f^\ell$, so our new steps of size ℓ now represent the movement of the entire cycle σ . Take one point x_i from σ . For $x \in x_i + p^k\mathbb{Z}_p$, define

$$a_k(x) := g'(x) = \prod_{j=0}^{\ell-1} f'(f^j(x)), \quad b_k(x) := \frac{g(x) - x}{p^k} = \frac{f^\ell(x) - x}{p^k}.$$

From the context of Hensel lifting, recall that a lift of x_i to level $k+1$ has the form $x_i + p^k t$, where $t \in \{0, \dots, p-1\}$. By the first-order Taylor expansion, or, equivalently, the *linearization* of g near x_i , we have

$$\begin{aligned} g(x_i + p^k t) &\equiv g(x_i) + p^k t g'(x_i) \pmod{p^{k+1}} \\ &\equiv x_i + p^k b_k(x_i) + p^k t a_k(x_i) \pmod{p^{k+1}} \\ &\equiv x_i + p^k (b_k(x_i) + a_k(x_i) t) \pmod{p^{k+1}}. \end{aligned}$$

so the behavior of the lifts of σ is now conveniently governed by the affine map

$$\Phi_{\sigma,i}(t) := b_k(x_i) + a_k(x_i) t \pmod{p}, \quad t \in \mathbb{Z}_p/p\mathbb{Z}_p.$$

Cycle lifting means replacing each point x_i of a cycle modulo p^k by its p lifts $x_i + p^k t$ modulo p^{k+1} ; the map $\Phi_{\sigma,i}$ tells us how these lifts move after one full turn around σ .

Definition 2.20. The cycle σ is said to *grow* if

$$a_k(x_i) \equiv 1 \pmod{p}, \quad b_k(x_i) \not\equiv 0 \pmod{p}.$$

Equivalently, $\Phi_{\sigma,i}(t) = t + b_k(x_i)$ is a single p -cycle on $\mathbb{Z}_p/p\mathbb{Z}_p$, so the lift of σ to level $k+1$ is one cycle of length ℓp .

Definition 2.21. The cycle σ is said to *split* if

$$a_k(x_i) \equiv 1 \pmod{p}, \quad b_k(x_i) \equiv 0 \pmod{p}.$$

Equivalently, $\Phi_{\sigma,i}(t) = t$ fixes every $t \in \mathbb{Z}_p/p\mathbb{Z}_p$, so the lift of σ to level $k+1$ splits into p cycles, each of length ℓ .

Definition 2.22. The cycle σ is said to *grow tails* if

$$a_k(x_i) \equiv 0 \pmod{p}.$$

Equivalently, $\Phi_{\sigma,i}$ is constant modulo p ; hence, among the lifts there is one distinguished cycle, and all other lifted residue classes are eventually attracted to it.

For the following proposition, let

$$U_k(\sigma) := \bigsqcup_{i=1}^{\ell} (x_i + p^k \mathbb{Z}_p)$$

denote the union of the residue balls attached to σ modulo p^k . Also, note that $f|_{U_k(\sigma)}$ means the restriction of f to $U_k(\sigma)$.

Proposition 2.23. [2, Theorem 2.3] Let σ be an ℓ -cycle of f_k . If σ grows at every higher level, then $f|_{U_k(\sigma)}$ is topologically conjugate to the adding machine with cycle lengths

$$\ell, \ell p, \ell p^2, \dots$$

In other words, on $U_k(\sigma)$, the map f is topologically equivalent to the adding-machine map $x \mapsto x + 1$, which means that each refinement of the cycle σ under f remains a single clean, uninterrupted cycle.

Lemma 2.24. [4] For odd primes, once a cycle grows at some level $n \geq 2$, it grows forever.

This is crucial for determining minimality; once a cycle σ at level k_0 grows forever by Lemma 2.24, every further reduction remains a single growing lift, so for any lifted cycle $\tilde{\sigma}$ at level $k \geq k_0$, we can definitively conclude minimality of the associated clopen invariant set $U_k(\tilde{\sigma})$.

3 Fermat Prime Dynamics of $x^2 - p^r x$ on $\mathbb{Z}_p$

Great progress in the study of dynamics on p -adic integers was made by S. Fan and L. Liao [7], who determined the minimal components of $(\mathbb{Z}_p, x^2)$. Their decomposition was strongly focused on structural and enumerative aspects and was not arranged as a single closed-form minimal decomposition with explicitly indexed p -adic balls. We build on this perspective by taking modified versions of the square maps of the form $x^2 - p^r x$ and studying them in the context of Fermat primes: a deliberate choice of a set of odd primes which substantially simplifies closed-form formulation. This method may also provide a useful foundation for algorithmic approaches in future research.

For the following subsection, let $(\mathbb{Z}_p, f)$ be a p -adic dynamical system where $f \in \mathbb{Z}_p[x]$, $f(x) = x^2 - p^r x$ with $r \in \mathbb{Z}^+$ and $p = 2^{2^\ell} + 1$ a Fermat prime. One notable finding that will quickly become apparent is that the case where $p = 5$ and $r = 1$ is exceptional and requires special attention. Note finally that we will sometimes use the alternative notation $\mathbb{F}_p^\times = (\mathbb{Z}_p/p\mathbb{Z}_p)^\times$ for brevity.

3.1 Minimal Components

Lemma 3.1. Minimal components are contained in the residue class $(1 + p^r) + p\mathbb{Z}_p = 1 + p\mathbb{Z}_p$, the ball centered at the indifferent fixed point $1 + p^r$.

Proof. We begin by finding fixed points. Setting $x^2 - p^r x = x$, we get $x^2 - x(1 + p^r) = x(x - (1 + p^r)) = 0$. Hence, the fixed points are $x = 0$ and $x = 1 + p^r$. To classify them, use the derivative $f'(x) = 2x - p^r$:

1. At $x = 0$, we have $f'(0) = -p^r$, so $|f'(0)|_p = p^{-r} < 1 \implies 0$ is an attracting fixed point.
2. At $x = 1 + p^r$, we have $f'(1 + p^r) = 2 + p^r$, so $|f'(1 + p^r)|_p = 1 \implies 1 + p^r$ is an indifferent fixed point.

Modulo p , the map reduces to x^2 . $0 \mapsto 0$, so (0) is a cycle. Now consider the rest: $|\mathbb{F}_p^\times| = p - 1 = 2^{2^\ell}$ so, by Lagrange's theorem (Theorem A.2), every nonzero residue $x \in \mathbb{F}_p^\times$ has order 2^m for some $m \geq 0$. Notice that if $x \neq 1$, $\text{ord}(x^2) = \frac{\text{ord}(x)}{2}$, so repeated squaring eventually leads to 1. Thus, all nonzero residues including 1 eventually map to 1, so the only cycles are (0) and (1) . Moreover, if $x \in p\mathbb{Z}_p$, then $|x|_p \leq p^{-1}$, and since $x, p^r \in p\mathbb{Z}_p$, for $x \neq 0$, we have

$$|f(x)|_p = |x|_p |x - p^r|_p \leq |x|_p p^{-1} < |x|_p.$$

Thus, $f^n(x) \rightarrow 0$, so $p\mathbb{Z}_p \subseteq \mathcal{B}(\{0\})$. By Theorem 2.15, the basin part $\mathcal{B}$ is disjoint from the minimal-component part $\mathcal{M}$. Therefore, $p\mathbb{Z}_p$ cannot contain minimal components.

Finally, by Theorem 2.17, the reduction of any minimal component modulo p must be minimal for f_1 , and so must also be a cycle of the finite map f_1 . Since the only cycles of $f_1(x) = x^2$ are (0) and (1) , and the class $p\mathbb{Z}_p$ has just been excluded, every minimal component must be contained in the remaining residue class $1 + p\mathbb{Z}_p = (1 + p^r) + p\mathbb{Z}_p$. $\square$

The following lemma is phrased in a way that deliberately supports the visual intuition. Recall that the balls associated with the reduction level k are of radius p^{-k} , so smaller reduction levels correspond to larger p -adic balls, and vice versa.

Lemma 3.2. The minimal components with the largest constituent ball radii arise from $2^{\ell+1}$ -cycles modulo p^3 if $p = 5$ and $r = 1$, and modulo p^2 otherwise.

Proof. From the proof of Lemma 3.1, we know that the only cycles modulo p are (0) and (1) . Neither cycle gives a minimal component at level p : the lift of (0) lies in the attracting basin of 0, while the lift of (1) contains the fixed point $1 + p^r$, whose orbit is not dense in $1 + p\mathbb{Z}_p$, making the full residue class non-minimal. Thus, minimal components must correspond to cycles appearing at reduction levels higher than 1. For $(p, r) \neq (5, 1)$, we show that such cycles of length $2^{\ell+1}$ appear modulo p^2 . In the exceptional case $(p, r) = (5, 1)$, we prove that the relevant cycle modulo p^2 splits, and that the minimal components corresponding to $2^{\ell+1}$ -cycles appear modulo p^3 instead.

First, consider the case $r = 1$ and work modulo p^2 . We have $x^2 - px \pmod{p^2}$. Take the residue class $1 \pmod{p^2}$, i.e., the ball $1 + p^2\mathbb{Z}_p$. We see that

$$f(1) = 1 - p, \quad f(1 - p) \equiv 1 - 3p \pmod{p^2}, \quad f(1 - 3p) \equiv 1 - 7p \pmod{p^2}, \quad \dots$$

meaning that

$$\begin{aligned} f(1 + p^2\mathbb{Z}_p) &\subseteq 1 - p + p^2\mathbb{Z}_p, \\ f(1 - p + p^2\mathbb{Z}_p) &\subseteq 1 - 3p + p^2\mathbb{Z}_p, \\ f(1 - 3p + p^2\mathbb{Z}_p) &\subseteq 1 - 7p + p^2\mathbb{Z}_p, \end{aligned}$$

and so on. This is because, in general, for $t \in \{0, \dots, p-1\}$,

$$\begin{aligned} f(1 + tp) &\equiv 1 + (2t - 1)p \pmod{p^2}, \\ f^j(1 + tp) &\equiv 1 + (2^j(t - 1) + 1)p \pmod{p^2}. \end{aligned}$$

In other words, after j iterations, the coefficient is $2^j(t - 1) + 1$. But since $p = 2^{2^\ell} + 1$, we have $2^{2^\ell} \equiv -1 \pmod{p}$, so $2^{2^{\ell+1}} \equiv 1 \pmod{p}$. Therefore, $\text{ord}_p(2) = 2^{\ell+1}$, so for $t \neq 1$, we have

$$f^{2^{\ell+1}}(1 + tp) \equiv 1 + tp \pmod{p^2},$$

meaning that after exactly $2^{\ell+1}$ steps, orbits return to $1 + tp \pmod{p^2}$, and so result in $2^{\ell+1}$ -cycles of the form

$$\sigma_t = \left(1 + tp, f(1 + tp), f^2(1 + tp), \dots, f^{2^{\ell+1}-1}(1 + tp) \right) \pmod{p^2}.$$

Now, take one such cycle σ_t . At the second reduction, compute $\Phi_{\sigma_t, 1+tp}$:

$$a_2(1 + tp) = \prod_{j=0}^{2^{\ell+1}-1} f'(f^j(1 + tp)), \quad b_2(1 + tp) = \frac{f^{2^{\ell+1}}(1 + tp) - (1 + tp)}{p^2}.$$

We expand $f^j(1 + tp)$ modulo p^3 as part of the calculation of b_2 ; set $u = t - 1$ for convenience:

$$f^j(1 + tp) \equiv 1 + (1 + 2^j u)p + 2^{j-1}(ju + (2^j - 1)u^2)p^2 \pmod{p^3}.$$

Thus, slightly suppressing the intermediate algebra,

$$\begin{aligned} b_2(1 + tp) &= \frac{f^{2^{\ell+1}}(1 + tp) - (1 + tp)}{p^2} \\ &\equiv \frac{(2^{2^{\ell+1}} - 1)u}{p} + 2^{2^{\ell+1}-1} \left(2^{\ell+1}u + (2^{2^{\ell+1}} - 1)u^2 \right) \\ &\equiv (2^\ell - 2)u \equiv (2^\ell - 2)(t - 1) \pmod{p}, \end{aligned}$$

while

$$a_2(1 + tp) = \prod_{j=0}^{2^{\ell+1}-1} f'(f^j(1 + tp)) \equiv 2^{2^{\ell+1}} \equiv 1 \pmod{p},$$

so

$$\Phi_{\sigma_t, 1+tp}(s) \equiv s + (2^\ell - 2)(t - 1) \pmod{p}.$$

If $p = 5$, then $\ell = 1$, so $b \equiv 0 \pmod{p}$. Hence, σ_t splits. In every other case, since $t \neq 1$, we have $t - 1 \not\equiv 0 \pmod{p}$. Also, $2^\ell - 2 \not\equiv 0 \pmod{p}$, since when $\ell = 0$, this expression equals -1 , and when $\ell \geq 2$, we have $0 < 2^\ell - 2 < p$. Therefore, $b \not\equiv 0 \pmod{p}$, so σ_t grows, and its minimality follows from Lemma 2.24.

For $r > 1$, the map reduces to $x^2 \pmod{p^2}$. We have

$$\begin{aligned} f(1 + tp) &\equiv (1 + tp)^2 \equiv 1 + 2tp \pmod{p^2}, \\ f^j(1 + tp) &\equiv 1 + 2^j tp \pmod{p^2}. \end{aligned}$$

For $t \not\equiv 0 \pmod{p}$, the return condition is $2^j \equiv 1 \pmod{p}$. Since $\text{ord}_p(2) = 2^{\ell+1}$, the orbit of $1 + tp$ modulo p^2 is a $2^{\ell+1}$ -cycle.

To compute b_2 , note that if $r \geq 3$, then $f(x) \equiv x^2 \pmod{p^3}$, so the calculation is exactly the same as for the squaring map. On the other hand, if $r = 2$, the term $-p^2x$ must be accounted for modulo p^3 , but after division by p^2 it gives a term proportional to $2^{2^{\ell+1}} - 1 \equiv 0 \pmod{p}$, and vanishes modulo p .

We can thus generalize that in all cases where $r > 1$,

$$\begin{aligned} b_2(1 + tp) &\equiv \frac{(1 + tp)^{2^{2^{\ell+1}}} - (1 + tp)}{p^2} \\ &\equiv t \frac{2^{2^{\ell+1}} - 1}{p} + \binom{2^{2^{\ell+1}}}{2} t^2 \\ &\equiv t(p - 2) + 0 \equiv -2t \pmod{p}, \\ a_2(1 + tp) &= \prod_{j=0}^{2^{\ell+1}-1} f'(f^j(1 + tp)) \equiv 2^{2^{\ell+1}} \equiv 1 \pmod{p}. \end{aligned}$$

This means that σ_t grows for any Fermat prime p , and so corresponds to a minimal component. Finally, let us return to the exceptional case $(p, r) = (5, 1)$. As we have determined earlier, the cycle

$$\sigma = (1, 21, 11, 16)$$

splits modulo 5^2 ; its five lifts modulo 5^3 are

$$\begin{aligned} \tilde{\sigma}_0 &= (1, 121, 36, 116), \\ \tilde{\sigma}_1 &= (26, 46, 11, 66), \\ \tilde{\sigma}_2 &= (51, 96, 111, 16), \\ \tilde{\sigma}_3 &= (76, 21, 86, 91), \\ \tilde{\sigma}_4 &= (101, 71, 61, 41). \end{aligned}$$

For the starting representatives $1 + 25j$, $0 \leq j \leq 4$, direct computation gives

$$b_3(1 + 25j) = \frac{f^4(1 + 25j) - (1 + 25j)}{5^3} \equiv 4 \pmod{5}.$$

Moreover, for each $\tilde{\sigma}_i$, every element is congruent to 1 (mod 5), so

$$a_3 \equiv 2^4 \equiv 1 \pmod{5}.$$

Thus, $\Phi_{\tilde{\sigma}_i, 1+25j}(s) \equiv s + 4 \pmod{5}$ for each i . Hence, all five cycles grow, and each corresponds to a minimal component by Lemma 2.24; on each such component, the dynamics is conjugate to an adding machine. $\square$

In the proof of the following lemma and beyond, we refer to cycles whose associated unions of clopen balls are minimal components as simply *minimal cycles* for succinctness.

Lemma 3.3. At reduction level k , the number of new minimal components is

$$\begin{cases} 2^{2^\ell - \ell - 1} & \text{if } (p, r) \neq (5, 1) \text{ and } k \geq 2, \\ 5 & \text{if } (p, r) = (5, 1) \text{ and } k \geq 3, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. From Lemma 3.2, we know that the first minimal cycles arise at reduction levels 2 and 3 for $(p, r) \neq (5, 1)$ and $(p, r) = (5, 1)$, respectively. Those cycles continue to grow for all further levels by Lemma 2.24. Thus, on each associated set $U_k(\sigma)$, every higher finite reduction remains a single cycle, and by Theorem 2.17, the restriction $f|_{U_k(\sigma)}$ is minimal. Note too that by Theorem 2.17 and Lemma 3.1, every minimal component must arise from the lifting of the cycle (1). Thus, the cycles at the levels identified in Lemma 3.2 account for all possible minimal cycles at every level.

For the non-exceptional case, the lifts of the residue class 1 modulo p are of the form $1 + tp$ modulo p^2 , where $t \in \{0, \dots, p-1\}$, which means that, after excluding the fixed point, there are $|\mathbb{F}_p^\times| = p-1$ balls in total. Since, as determined in Lemma 3.2, the minimal cycles are of lengths $\text{ord}_p(2) = 2^{\ell+1}$, we can obtain the initial number of minimal cycles by computing

$$\frac{|\mathbb{F}_p^\times|}{\text{ord}_p(2)} = \frac{p-1}{\text{ord}_p(2)} = \frac{2^{2^\ell}}{2^{\ell+1}} = 2^{2^\ell - \ell - 1}.$$

Notice that each of those minimal cycles ends up with lengths multiplied by p after lifting. Also, let $m_k = \text{ord}_{p^k}(2)$. If $2^{m_k} = 1 + bp^k$ with $p \nmid b$, then

$$2^{am_k} \equiv 1 + abp^k \not\equiv 1 \pmod{p^{k+1}} \quad (1 \leq a < p), \quad 2^{pm_k} \equiv 1 \pmod{p^{k+1}},$$

so $\text{ord}_{p^{k+1}}(2) = p \text{ord}_{p^k}(2)$, which means that, at each higher reduction, each growing cycle lifts to a single cycle whose length is multiplied by p . The number of nonzero lifts at the next level is also multiplied by p , so the quotient giving the number of new minimal cycles remains unchanged. Thus, the number of new minimal components stays fixed for reductions $k \geq 2$.

In the exceptional case, the only cycles modulo 5^2 are (0), (6), and the splitting cycle (1, 21, 11, 16). As we have seen in the proof of Lemma 3.2, modulo

5^3 , the lifts of the splitting cycle become 5 minimal components and grow forever by Lemma 2.24.

The lifts of the fixed point 6 modulo 5^k to modulo 5^{k+1} are

$$6, \quad 6 + 5^k, \quad 6 + 2 \cdot 5^k, \quad 6 + 3 \cdot 5^k, \quad 6 + 4 \cdot 5^k.$$

We see that $f(6) = 6$, so the fixed point stays fixed. For $u \in \{1, 2, 3, 4\}$, expand

$$\begin{aligned} f(6 + u5^k) &= (6 + u5^k)^2 - 5(6 + u5^k) \\ &= 36 + 12u5^k + u^25^{2k} - 30 - u5^{k+1} \\ &\equiv 6 + 2u5^k \pmod{5^{k+1}}, \end{aligned}$$

so the values get multiplied by 2 after passing through f , and thus form a 4-cycle. Let $x = 6 + u5^k$, then

$$\begin{aligned} a_{k+1}(x) &= (f^4)'(x) = \prod_{i=0}^3 f'(f^i(x)) = \prod_{i=0}^3 (2f^i(x) - 5) \equiv 2^4 \equiv 1 \pmod{5}, \\ b_{k+1}(x) &= \frac{f^4(x) - x}{5^{k+1}} \equiv \frac{(7^4 - 1)u5^k}{5^{k+1}} = 480u \equiv 0 \pmod{5}. \end{aligned}$$

Thus, the 4-cycle splits.

The five lifts are then $y = 6 + u5^k + m5^{k+1}$, where $m \in \{0, 1, 2, 3, 4\}$. We have

$$\begin{aligned} a_{k+2}(y) &= (f^4)'(y) = \prod_{i=0}^3 f'(f^i(y)) = \prod_{i=0}^3 (2f^i(y) - 5) \equiv 2^4 \equiv 1 \pmod{5}, \\ b_{k+2}(y) &= \frac{f^4(y) - y}{5^{k+2}} \equiv \frac{(7^4 - 1)(u5^k + m5^{k+1})}{5^{k+2}} \equiv 96u \equiv u \not\equiv 0 \pmod{5}. \end{aligned}$$

Thus, the lifts grow.

The preceding reduction calculation shows that (1) gives rise only to splitting cycles, which are not minimal components, while those splitting cycles give rise to minimal cycles. Thus, the only minimal components at the relevant reductions are the lifts of the splitting cycle from the preceding reduction, and there are precisely five of them. $\square$

The mechanism is therefore repetitive. In particular, in the exceptional case, once the entire “blueprint” of the dynamics is visible modulo 5^3 , we can make generalizations about the entire system’s behavior.

3.2 Closed-Form Minimal Decomposition

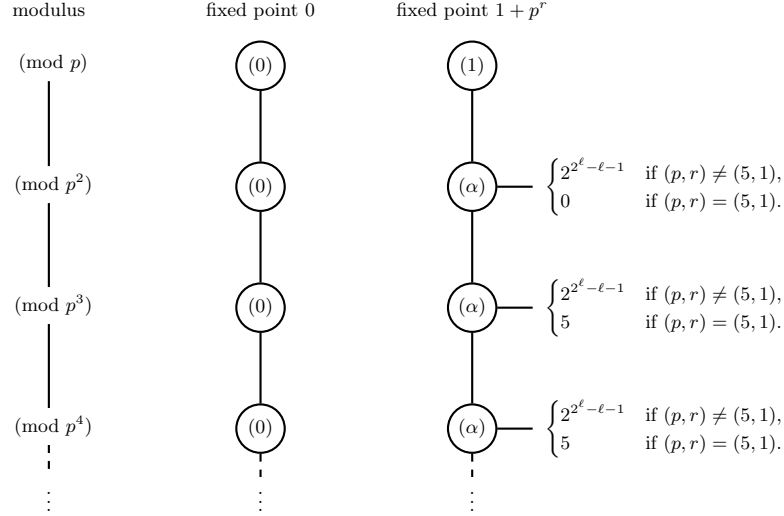


Figure 2. Partial reduction-level structure and number of new minimal components for $f(x) = x^2 - p^r x$ over $\mathbb{Z}_p$, where $p = 2^{2^\ell} + 1$ and $\alpha = 1 + p^r$

Note that the diagram shows only the cycles relevant to the decomposition and is not fully comprehensive.

We now use our knowledge about the behavior of the system to give the generalized minimal decomposition. In the theorem below, $\mathbb{F}_p^\times / \langle 2 \rangle$ denotes the nonzero residue classes modulo p grouped into orbits under multiplication by powers of 2. Concretely, after choosing a representative $t \in \{1, \dots, p-1\}$, the class $[t]$ is

$$[t] = \{t, 2t, \dots, 2^{2^{\ell+1}-1}t\} \pmod{p}.$$

Importantly, $|\langle 2 \rangle| = \text{ord}_p(2) = 2^{\ell+1}$, so

$$|\mathbb{F}_p^\times / \langle 2 \rangle| = \frac{|\mathbb{F}_p^\times|}{|\langle 2 \rangle|} = 2^{2^\ell - \ell - 1}.$$

Theorem 3.4. Let $(\mathbb{Z}_p, f)$ be a p -adic dynamical system with $f(x) = x^2 - p^r x \in \mathbb{Z}_p[x]$, where $r \in \mathbb{Z}^+$ ($r \neq 1$ if $p = 5$), and $p = 2^{2^\ell} + 1$ is a Fermat prime. Let $\alpha = 1 + p^r$. Then $\mathbb{Z}_p$ admits the following minimal decomposition:

$$\begin{aligned} \mathbb{Z}_p = \{0, \alpha\} \sqcup \bigsqcup_{k \geq 1} \bigsqcup_{[t] \in \mathbb{F}_p^\times / \langle 2 \rangle} \bigsqcup_{j=0}^{2^{\ell+1}-1} (\alpha + 2^j t p^k + p^{k+1} \mathbb{Z}_p) \\ \sqcup (\mathbb{Z}_p \setminus (\{0\} \cup (1 + p\mathbb{Z}_p))). \end{aligned}$$

Proof. Theorem 2.15 says that a full minimal decomposition of the form

$$\mathbb{Z}_p = \mathcal{P} \sqcup \mathcal{M} \sqcup \mathcal{B}$$

exists, and we are trying to show that these disjoint parts are exactly

$$\begin{aligned} \mathcal{P} &= \{0, \alpha\}, \\ \mathcal{M} &= \bigsqcup_{k \geq 1} \bigsqcup_{[t] \in \mathbb{F}_p^\times / \langle 2 \rangle} \bigsqcup_{j=0}^{2^{\ell+1}-1} (\alpha + 2^j t p^k + p^{k+1} \mathbb{Z}_p), \\ \mathcal{B} &= (\mathbb{Z}_p \setminus (\{0\} \cup (1 + p\mathbb{Z}_p))). \end{aligned}$$

In the proof of Lemma 3.1, we found the fixed points 0 and α . Since $p\mathbb{Z}_p$ is attracted to 0, the only periodic point in $p\mathbb{Z}_p$ is 0. Thus, to verify the decomposition, it suffices to show that $\mathcal{P} = \{0, \alpha\}$ contains all the periodic points, and that $\mathcal{M}$ contains all the minimal components.

By Lemma 3.2, the first minimal components emerge modulo p^2 , which is consistent with our formulation. Following the same repeated-squaring mechanism used in the proof of Lemma 3.1, the action on balls of the form $\alpha + tp^k + p^{k+1}\mathbb{Z}_p$ sends t to $2t$ modulo p . Therefore, the balls are naturally grouped into $\langle 2 \rangle$ -orbits, represented by the classes $[t] \in \mathbb{F}_p^\times / \langle 2 \rangle$. For this reason, choosing a single representative from each doubling orbit is enough. Since $|\mathbb{F}_p^\times / \langle 2 \rangle| = 2^{2^\ell - \ell - 1}$ as established earlier, our logic agrees with Lemma 3.3.

In the case $r = 1$, the coefficient $2^j(s-1) + 1$ from the proof of Lemma 3.2 becomes $2^j(s-1)$ after centering the balls at $\alpha = 1 + p$. Thus, replacing $s-1$ by $t \in \mathbb{F}_p^\times$, the balls are indexed by $2^j t$. For $r > 1$, the same proof already gives the coefficient $2^j s$, since $f(x) \equiv x^2 \pmod{p^2}$ and $\alpha \equiv 1 \pmod{p^2}$, so the same indexing by $2^j t$ applies.

The minimality follows by the same growth calculation used in the proof of Lemma 3.2. By the preceding lifting analysis, the displayed set $\mathcal{M}$ is exactly the union of all minimal components: it contains precisely $2^{2^\ell - \ell - 1}$ cycles at each level, and each such cycle corresponds to a minimal component. We have also identified the only periodic points, namely 0 and α . Therefore, $\mathcal{P} = \{0, \alpha\}$ is correct, and since the remaining part is just $\mathcal{B} = \mathbb{Z}_p \setminus (\mathcal{P} \sqcup \mathcal{M})$, no separate verification of $\mathcal{B}$ is required. Thus, the stated decomposition is correct. $\square$

Corollary 3.5. In the non-exceptional case, every nonzero residue inside $1 + p\mathbb{Z}_p$ is part of a minimal component.

Proof. This follows immediately from Theorem 3.4. $\square$

Another intuitive way to think about the triple disjoint union is as a structure of three nested loops in a computer algorithm. First, we iterate over reductions; second, over the minimal components; third, over the underlying p -adic balls. For each $k \geq 1$, the balls have radius $p^{-(k+1)}$, so $k+1$ is the reduction level. At each such level, the class $[t] \in \mathbb{F}_p^\times / \langle 2 \rangle$ chooses one minimal component, while $j = 0, \dots, 2^{\ell+1} - 1$ runs through the balls contained in that component.

3.3 The Exceptional Case

We now state the minimal decomposition of the exceptional case where $p = 5$ and $r = 1$.

Theorem 3.6. The 5-adic dynamical system $(\mathbb{Z}_5, x^2 - 5x)$ admits the following minimal decomposition:

$$\begin{aligned} \mathbb{Z}_5 = \{0, 6\} \sqcup \bigsqcup_{k \geq 2} \bigsqcup_{m=1}^4 \bigsqcup_{j=0}^4 (6 + 5^{k-1}m + 5^k j + 5^{k+1}\mathbb{Z}_5) \\ \sqcup (\mathbb{Z}_5 \setminus (\{0\} \cup (6 + 5\mathbb{Z}_5))). \end{aligned}$$

Proof. Following the reasoning used in the proof of Theorem 3.4, it suffices to show that the triple disjoint union, which here represents the minimal part $\mathcal{M}$, contains all the minimal components and nothing but minimal components. We start from reduction $k + 1 = 3$, which agrees with Lemma 3.2. From the proof of Lemma 3.3, we know that the fixed point 6 gives rise to a splitting cycle, and that the splitting cycle lifts to precisely five growing 4-cycles.

We write $6 + 5^{k-1}m$, with $m \in \{1, 2, 3, 4\}$, for the elements of the splitting cycle modulo 5^k , and add $5^k j$, with $j \in \{0, 1, 2, 3, 4\}$, to range over their lifts modulo 5^{k+1} , since every lift of a residue class modulo 5^k is obtained uniquely in this way. Thus, we have five (consistent with Lemma 3.3) 4-cycles for each relevant reduction, and each of them corresponds to a minimal component, which was calculated in the proof of Lemma 3.3. Therefore, the minimal part $\mathcal{M}$ is correct, and so is the entire decomposition. $\square$

Remark 3.7. In the decomposition given in Theorem 3.6, the minimal part

$$\mathcal{M} = \bigsqcup_{k \geq 2} \bigsqcup_{m=1}^4 \bigsqcup_{j=0}^4 (6 + 5^{k-1}m + 5^k j + 5^{k+1}\mathbb{Z}_5)$$

is not written as a union of explicit minimal components $\mathcal{M}_i$, unlike in the non-exceptional decomposition given in Theorem 3.4.

There are many different ways to state the same decomposition. For example, one could make use of predefined cycles, and even use matrices for structured grouping thereof. Remark 3.7 points out one of the reasons for choosing alternative notation: this would allow, for instance, the minimal components to be written out explicitly.

4 Examples

We consider two examples: first, with $(\mathbb{Z}_3, x^2 - 3x)$, and second, with $(\mathbb{Z}_{257}, x^2 - 257^3 x)$. Our goal here is to state minimal decompositions using Theorem 3.4, explicitly tracking fixed and periodic points, cycles, reductions modulo powers of p , and the induced action of the polynomial maps on p -adic balls at the

same time. Since the exceptional case $p = 5$ and $r = 1$ is treated separately in Theorem 3.6, we focus here on examples with primes $p \neq 5$. This is only for variation; Theorem 3.4, of course, still applies to $p = 5$ whenever $r > 1$.

4.1 The 3-adic Case

Take the 3-adic dynamical system $(\mathbb{Z}_3, x^2 - 3x)$. By defining an auxiliary function $g(x) := f(x) - x$ and solving $g(x) = 0$, we obtain the fixed points $x = 0$ and $x = 4$. To classify them, just like in the proof of Lemma 3.1, compute $f'(x) = 2x - 3$.

1. At $x = 0$, we have $f'(0) = -3$, so $|f'(0)|_3 = 1/3 < 1 \implies 0$ is an attracting fixed point.
2. At $x = 4$, we have $f'(4) = 5$, so $|f'(4)|_3 = 1 \implies 4$ is an indifferent fixed point.

Calculate the reduction mod 3: $f_1(x) \equiv x^2 - 3x \equiv x^2 \pmod{3}$.

$$0 \mapsto 0, \quad 1 \mapsto 1, \quad 2 \mapsto 1.$$

Note that by Theorem 2.18, the only possible periods are 1, 2, 3, 4, 6, and 9. The observed 1-cycles are thus consistent with this restriction.

Next, we have $f_2(x) \equiv x^2 - 3x \pmod{9}$. As the reduction k increases, the residue-level dynamics are most conveniently represented diagrammatically; we observe the following behavior.

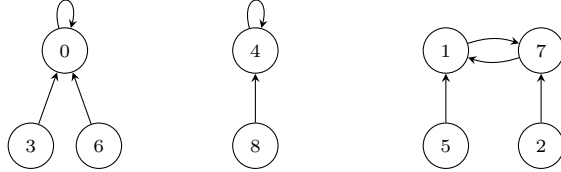


Figure 3. Dynamics of $f(x) = x^2 - 3x \in \mathbb{Z}_3[x]$ modulo 9.

The resulting cycles are (0) , (4) , and $(1, 7)$, which is expected and reveals both of the fixed points. Our third reduction is $f_3(x) \equiv x^2 - 3x \pmod{27}$, and leads to the following structure.

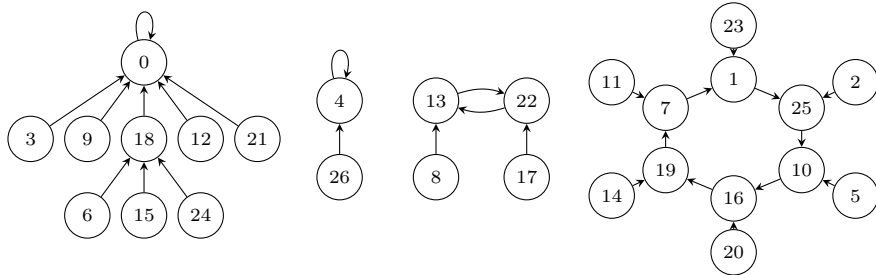


Figure 4. Dynamics of $f(x) = x^2 - 3x \in \mathbb{Z}_3[x]$ modulo 27.

Theorem 2.18 indicates that the observed 6-cycle $(1, 25, 10, 16, 19, 7)$ in this reduction is of second-largest attainable length. Crucially, the diagram is not merely a finite dynamical system on $\mathbb{Z}/27\mathbb{Z}$, but also a rough picture of the induced action of f on the partition

$$\mathbb{Z}_3 = \bigsqcup_{a=0}^{26} (a + 27\mathbb{Z}_3).$$

For example, the 6-cycle $\sigma = (1, 25, 10, 16, 19, 7)$ modulo 27 determines the f -invariant clopen set

$$\begin{aligned} U_3(\sigma) &= (1 + 27\mathbb{Z}_3) \cup (25 + 27\mathbb{Z}_3) \cup (10 + 27\mathbb{Z}_3) \\ &\quad \cup (16 + 27\mathbb{Z}_3) \cup (19 + 27\mathbb{Z}_3) \cup (7 + 27\mathbb{Z}_3). \end{aligned}$$

In general, recall that p -adic ball structure can be written as

$$a + p^k\mathbb{Z}_p = \bigsqcup_{r=0}^{p-1} (a + rp^k + p^{k+1}\mathbb{Z}_p).$$

In particular, for $p = 3$, this becomes

$$a + 3^k\mathbb{Z}_3 = (a + 3^{k+1}\mathbb{Z}_3) \sqcup (a + 3^k + 3^{k+1}\mathbb{Z}_3) \sqcup (a + 2 \cdot 3^k + 3^{k+1}\mathbb{Z}_3).$$

Before stating the decomposition, we finally note that

$$\begin{aligned} f(3\mathbb{Z}_3) &\subseteq 3\mathbb{Z}_3 &\implies 3\mathbb{Z}_3 \text{ is } f\text{-invariant,} \\ f(1 + 3\mathbb{Z}_3) &\subseteq 1 + 3\mathbb{Z}_3 &\implies 1 + 3\mathbb{Z}_3 \text{ is } f\text{-invariant,} \\ f(2 + 3\mathbb{Z}_3) &\subseteq 1 + 3\mathbb{Z}_3 &\implies 2 + 3\mathbb{Z}_3 \text{ is not } f\text{-invariant.} \end{aligned}$$

This is important because, even before knowing the full decomposition, these inclusions already indicate that any nontrivial minimal dynamics must come from the invariant ball $1 + 3\mathbb{Z}_3$; of course, we can also use Lemma 3.1, but, even without it, we observe that the class $2 + 3\mathbb{Z}_3$ is sent into $1 + 3\mathbb{Z}_3$ after one iterate, while $3\mathbb{Z}_3$ belongs to the attracting basin of 0.

Proposition 4.1. The 3-adic dynamical system $(\mathbb{Z}_3, f)$ induced by the polynomial $f(x) = x^2 - 3x$ admits the following minimal decomposition:

$$\begin{aligned} \mathbb{Z}_3 &= \{0, 4\} \sqcup \bigsqcup_{k \geq 1} [(4 + 3^k + 3^{k+1}\mathbb{Z}_3) \sqcup (4 + 2 \cdot 3^k + 3^{k+1}\mathbb{Z}_3)] \\ &\quad \sqcup ((3\mathbb{Z}_3 \setminus \{0\}) \sqcup (2 + 3\mathbb{Z}_3)). \end{aligned}$$

Here, $\{0, 4\}$ is the fixed point set, each two-ball union in the middle term is an f -invariant clopen minimal component, and every point in the final term lies in the basin of attraction of either a periodic orbit or a minimal subsystem.

Proof. This is the specialization of Theorem 3.4 to the values $p = 3$ and $r = 1$. $\square$

In this case, the only doubling orbit in $\mathbb{F}_3^\times$ is $\{1, 2\}$, so, when applying Theorem 3.4, it is enough to take the single representative $t = 1$, with $j = 0, 1$. Also, notice that the term containing the minimal components can alternatively be stated as

$$\mathcal{M} = \bigsqcup_{k \geq 1} [(4 + 3^k \mathbb{Z}_3) \setminus (4 + 3^{k+1} \mathbb{Z}_3)] = (1 + 3\mathbb{Z}_3) \setminus \{4\},$$

where the final equality follows from Corollary 3.5. While this is equivalent and noticeably more compact, the original form displays the minimal components more explicitly.

The first few layers of the minimal part $\mathcal{M}$ of the expression are thus

$$\begin{aligned} k = 1 : & \quad (7 + 9\mathbb{Z}_3) \sqcup (10 + 9\mathbb{Z}_3), \\ k = 2 : & \quad (13 + 27\mathbb{Z}_3) \sqcup (22 + 27\mathbb{Z}_3), \\ k = 3 : & \quad (31 + 81\mathbb{Z}_3) \sqcup (58 + 81\mathbb{Z}_3). \end{aligned}$$

4.2 The 257-adic Case

We now consider our last example, the system $(\mathbb{Z}_{257}, x^2 - 257^3 x)$. We treat it less explicitly than the preceding 3-adic example, since the main ideas have already been illustrated there and the 257-adic structure is much harder to visualize.

By Lemma 3.2, we may begin from the second reduction. We have

$$f_2(x) = x^2 - 257^3 x \equiv x^2 \pmod{257^2}.$$

It is clear that the cycles corresponding to fixed points are (0) and (1). Since $\text{ord}_{257}(2) = 16$, the nonzero residues split into $256/16 = 16$ doubling orbits, giving precisely 16 minimal components, each formed by a cycle of length 16, which agrees with Lemma 3.2 and Lemma 3.3. Choosing one representative from each orbit gives

$$t \in \{1, 3, 5, 7, 9, 11, 13, 15, 19, 21, 23, 25, 27, 37, 43, 45\}.$$

Now we have everything we need to apply Theorem 3.4.

Proposition 4.2. The 257-adic dynamical system $(\mathbb{Z}_{257}, f)$ induced by the polynomial $f(x) = x^2 - 257^3 x$ admits the following minimal decomposition:

$$\begin{aligned} \mathbb{Z}_{257} = \{0, \alpha\} \sqcup \bigsqcup_{k \geq 1} \bigsqcup_{t \in R} \bigsqcup_{j=0}^{15} [(\alpha + 2^j \cdot 257^k \cdot t + 257^{k+1} \mathbb{Z}_{257})] \\ \sqcup (\mathbb{Z}_{257} \setminus (\{0\} \cup (1 + 257\mathbb{Z}_{257}))), \end{aligned}$$

where $\alpha = 1 + 257^3$ and $R = \{1, 3, 5, 7, 9, 11, 13, 15, 19, 21, 23, 25, 27, 37, 43, 45\}$.

Proof. This is the specialization of Theorem 3.4 to the values $p = 257$ and $r = 3$. $\square$

The minimal components at the second reduction are as follows:

$$M_t = \bigsqcup_{j=0}^{15} (1 + 2^j \cdot 257t + 257^2 \mathbb{Z}_{257}), \quad t \in R.$$

For example,

$$\begin{aligned} M_1 &= (1 + 257 \mathbb{Z}_{257}) \sqcup (1 + 2 \cdot 257 + 257^2 \mathbb{Z}_{257}) \\ &\quad \sqcup (1 + 4 \cdot 257 + 257^2 \mathbb{Z}_{257}) \sqcup \cdots, \\ M_3 &= (1 + 3 \cdot 257 + 257^2 \mathbb{Z}_{257}) \sqcup (1 + 6 \cdot 257 + 257^2 \mathbb{Z}_{257}) \\ &\quad \sqcup (1 + 12 \cdot 257 + 257^2 \mathbb{Z}_{257}) \sqcup \cdots, \\ M_5 &= (1 + 5 \cdot 257 + 257^2 \mathbb{Z}_{257}) \sqcup (1 + 10 \cdot 257 + 257^2 \mathbb{Z}_{257}) \\ &\quad \sqcup (1 + 20 \cdot 257 + 257^2 \mathbb{Z}_{257}) \sqcup \cdots. \end{aligned}$$

A Auxiliary Results

We record the following standard results without proof.

Lemma A.1 (Hensel's lemma). Let $F \in \mathbb{Z}_p[x]$, and suppose $a \in \mathbb{Z}_p$ satisfies

$$F(a) \equiv 0 \pmod{p} \quad \text{and} \quad F'(a) \not\equiv 0 \pmod{p}.$$

Then there exists a unique $\alpha \in \mathbb{Z}_p$ such that

$$\alpha \equiv a \pmod{p} \quad \text{and} \quad F(\alpha) = 0.$$

Theorem A.2 (Lagrange's theorem). Let G be a finite group, and let $H \leq G$. Then $|H|$ divides $|G|$. In particular, for every $g \in G$, the order of g divides $|G|$.

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